

EE 314: Digital Signal Processing

Course Code	EE 314
Course Name	Digital Signal Processing
L-T-P-C	3-0-2-4
Prerequisites	IC210: Probability, Statistics and Random Processes, EE 304: Communication Theory or the instructor's consent
Intended for	B.Tech. 2nd, 3rd year students
Mutual Exclusion	EEXXX (Signals & Systems)
Approval	54th BoA (May 2024)
Recoded from	EE 571 by the 55th BoA (May 2024)

Revision:

Course Contents

- **Review of discrete-time signals and systems:** LTI systems and their properties, impulse response and convolution, Difference equations, Eigen-functions of LTI systems. (4 hours)
- **Fourier and Z-transforms:** Discrete-time Fourier transform and its properties; Z-transform and its properties; Discrete Fourier Transform and its properties; Fast Fourier Transform. (10 hours)
- **Sampling:** Time domain and frequency domain representation, Nyquist theorem, Signal reconstruction, Discrete-time processing of continuous-time signals, Continuous-time processing of discrete-time signals, Changing the sampling rate, Multi-rate signal processing, Sub-Nyquist sampling and its applications. (8 hours)
- **Filtering:** Discrete-time frequency selective filtering, Phase distortion and delay, Characterization with difference equations, Stability and Causality, Frequency response of rational system functions, All pass and minimum phase systems, Basics of filter design, Z-transform characterization of IIR filters, Window functions for FIR filters, Filter structures for IIR and FIR filters. (10 hours)
- **Introduction to wavelets and their applications:** Haar expansions, Wavelets in continuous time, Discrete Wavelet Transform using Haar basis, Construction of wavelets using Fourier techniques, Multiresolution analysis and construction of the wavelet. (10 hours)

Laboratory Modules

- The exercises/projects should cover various real-world applications of DSP. A few suggested topics: image processing and compression; Speech processing; Compressed sensing; Signal processing in communication systems

Text Books

1. Alan V. Oppenheim, Ronald W. Schaffer, John R. Buck., **Discrete-Time Signal Processing**, 2nd edition, Pearson, 1999.
2. Martin Vetterli, Jelena Kovacevic and Vivek Goyal, **Fourier and Wavelet Signal Processing**, Cambridge University Press, 2014

References

1. John G. Proakis, Dimitris G. Manolakis., **Digital Signal Processing – Principles, Algorithms, and Applications**, 4th Edition, Pearson 2007.
2. Stephen Mallat, **A Wavelet Tour of Signal Processing The Sparse Way**, Elsevier, 2009